simultaneously for scaling and resilience, that would be a problem. If you have numerous Dispatcher Service instances, you don't want an order to be dispatched from all of them. That would lead to errors and an inconsistent state.

Consumer groups solve the problem. All consumers in the same group share a single subscription. As a consequence, each message arriving at the queue to which they're subscribed will be processed by one consumer only. Assume we have two applications (Dispatcher Service and Mail Service) interested in receiving events about accepted orders and deployed in a replicated fashion. Using the application name to configure consumer groups, we can ensure that each event is received and processed by a single instance of Dispatcher Service and a single instance of Mail Service, as shown in figure 10.11.

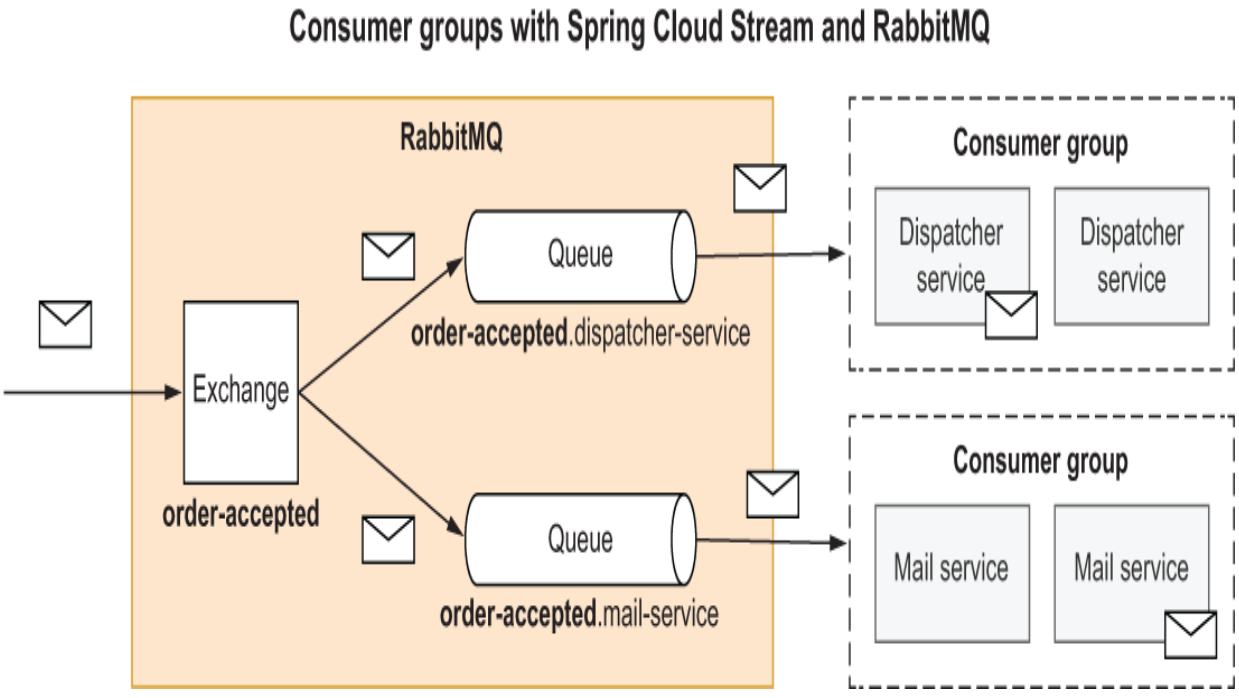


Figure 10.11 Consumer groups ensure that each message is received and processed by only one consumer within the same group.